

1. In class we have seen that the representations of the Lorentz group can be defined on the basis states $|j_1 m_1, j_2 m_2\rangle$, where

$$(J^A)^2 |j_1 m_1, j_2 m_2\rangle = j_1(j_1 + 1) |j_1 m_1, j_2 m_2\rangle \quad (1)$$

$$(J^B)^2 |j_1 m_1, j_2 m_2\rangle = j_2(j_2 + 1) |j_1 m_1, j_2 m_2\rangle \quad (2)$$

and the matrices J_i^A and J_i^B satisfy the usual angular momentum algebra. Recall that the rotation and boost generators are related to these by $\hat{L}_i = (J_i^A + J_i^B)$ and $\hat{K}_i = -i(J_i^B - J_i^A)$. The corresponding irreducible representations are labeled as $D^{j_1, j_2}(\Lambda)$.

Which finite dimensional representation (j_1, j_2) is equivalent to the matrices Λ_ν^μ as they operate on the space-time 4 vector x^μ ? Find explicitly the combination of basis states $|j_1 m_1, j_2 m_2\rangle$ which transform like the components x, y, z and t of the 4 vector.

2. Consider the extension of the proper Lorentz group to include space reflection or parity operation P defined by $P_\nu^\mu x^\nu = (t, -x, -y, -z)$.
- (a) Show that P commutes with the rotation generators J_i and anticommutes with the boost generators K_i .
- (b) Show that there is no 2×2 matrix with the properties of P within the $(1/2, 0)$ representation of the Lorentz group. Therefore the $(1/2, 0)$ representation of the Lorentz group cannot be extended to include parity.
3. Consider the electric and magnetic fields $\vec{E}$ and $\vec{B}$. Show that the combinations $\vec{E} + i\vec{B}$ and $\vec{E} - i\vec{B}$ transform amongst themselves under the Lorentz transformations. You may pick just one boost and one rotation to explicitly show this part. What happens to these combinations under space inversion? Under which representation of the Lorentz group do these combinations transform?

4. The Poincare group is obtained by combining proper Lorentz transformations and translations. A general element can be denoted by the pair $L = (a, \Lambda)$, where a^μ is the displacement and Λ^μ_ν a Lorentz transformation. Under the combined transformation

$$x^\mu \rightarrow x'^\mu = (Lx)^\mu = \Lambda^\mu_\nu x^\nu + a^\mu \quad (3)$$

Given the elements $L_1 = (a_1, \Lambda_1)$ and $L_2 = (a_2, \Lambda_2)$, show that the product element can be represented as $L_2 L_1 = (a_2 + \Lambda_2 a_1, \Lambda_2 \Lambda_1)$.